TOPIC 2 · PERSONALISATION, .RULES AND SKILLS FOR EXCEL
LAB 4 THEME · WORKBOOK .RULES

# Lab 4 — Create and Test Workbook .Rules for Water Service Reporting

| | |
|---|---|
| **Maps to** | LO2: distinguish account Personalisation, workbook .Rules and reusable Skills; configure, apply and verify each using water-service evidence. |
| **Duration** | 20 minutes |
| **Tools** | Microsoft 365 Copilot in Excel |
| **Mock data** | data/Northstar_Work_Orders.xlsx |
| **Course** | GenAI for Engineers · v2.4 |
| **Prompts** | https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-4 |

## Goal

Give a workbook its own rules with a .Rules sheet, then see Copilot follow them without being told.

## What You Will Build

A workbook with a visible .Rules sheet and a Rules_Test sheet that follows it.

## Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac. Download this lab folder and open the files straight from its data folder; save any new file in the same folder. All data is fictional.

**Data note**

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

## Steps

**Step 1.** Add the rules sheet. Open data/Northstar_Work_Orders.xlsx in Excel. In Copilot choose Allow editing, send the prompt and select Done. A sheet named .Rules appears as the last tab, with one rule in each cell of column A.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - ALLOW EDITING:
Add a visible worksheet named exactly .Rules (with the leading dot) as the last tab; if it
exists, clear column A and reuse it. Fill column A only, one cell per line below, entering
only the text after each "A#: " label. Do not change any other sheet.
A1: ## SOURCE AND SCOPE
A2: Preserve Work_Orders, tblWorkOrders, Technicians, current Assigned_To values and source
IDs.
A3: Use the reporting date from Control!B2. State the selected Site, Job_Category, Status and
Assigned_To scope beside every report.
A4: Backlog excludes Closed and Cancelled. Overdue backlog has Due_Date strictly before
Control!B2.
A5: On-time closure rate uses Closed rows only, with Closed_Date <= Due_Date. Display N/A when
no Closed rows remain.
A6: ## FORMATS AND VISUALS
A7: Display recorded Cost_SGD as S$ with two decimals; keep underlying values exact. Show
dates as DD-MMM-YYYY.
```

```
A8: Use white backgrounds, navy titles, blue and teal accents, readable labels and named units; never use colour alone.
A9: ## DASHBOARD LAYOUT
A10: Dashboard sheet: title in A1; Site, Job_Category, Status and Assigned_To dropdowns in B2:B5 (All plus source values); the six measures in B7:B12; data-as-at date and selected scope in D2.
A11: Six KPI tiles in D4:I7, one column per tile: title in row 4, large value in row 5, short definition in row 6 and "Checks: PASS" in row 7.
A12: Charts in a grid below the tiles: work orders by status and by job category side by side, then a full-width stacked bar by site (overdue backlog, other backlog, Closed, Cancelled).
A13: Slicers for Site, Job_Category, Status and Assigned_To on the right of the tiles, connected to tblWorkOrders. KPI formulas use SUBTOTAL visibility so slicers and dropdowns both filter every tile and chart.
A14: Put chart summaries, chart-title formulas and KPI checks on Dashboard_Summaries. Never merge cells. Hide gridlines on Dashboard.
A15: ## DASHBOARD STYLE
A16: Tiles: white bold text on solid navy #1F4E79 (totals and rates), blue #2E75B6 (backlog, overdue) or teal #2A9D8F (Closed, cost).
A17: Every chart title states the finding from a cell formula, e.g. "Changi Workshop has the most overdue backlog (18 of 50)".
A18: Show data labels on bars; no gridlines; rates as % with one decimal. Each chart total must equal B7.
A19: ## OUTPUT AND REVIEW
A20: Use structured table references. Preserve the visible .Rules sheet and unrelated sheets, charts and Prompt_Log.
A21: Keep current owners separate from unapproved assignment proposals. Missing facts and approvals remain UNKNOWN.
A22: Ask before replacing named output sheets. Reconcile every count, rate and chart against source rows; do not duplicate outputs on rerun.
```

**Step 2.** Add a date rule yourself. On the .Rules sheet click the first empty cell in column A, type the formula below and press Enter. It shows Reporting date: 2026-09-30.

**PROMPT / CHECK — copy and adapt**

```
="Reporting date: "&TEXT(Control!B2,"yyyy-mm-dd")
```

**Step 3.** Test the rules. Select New chat, choose Allow editing and send the prompt, then select Done. The prompt does not mention formats, but Rules_Test follows the rules: 120 total, 60 backlog, 50 overdue, on-time 30/48 = 62.5%, cost S$35,827.50.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - ALLOW EDITING:
Add a sheet named Rules_Test with the total work orders, backlog, overdue backlog, on-time closure rate and total cost from tblWorkOrders, using formulas.
```

**Step 4.** Check the rules travel with the file. Save, then use File > Save As to save a copy named Northstar_Rules_Copy.xlsx. In the copy choose Chat only and send the prompt: Copilot lists the same rules.

**PROMPT / CHECK — copy and adapt**

```
EXCEL - CHAT ONLY:
What rules does this workbook give you? List them briefly.
```

**Copy buttons for every prompt:** https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-4

## Test It

| Check | Expected result |
|---|---|
| .Rules sheet | Named exactly .Rules, visible, one rule per cell in column A. |
| Rules followed | Rules_Test: 120 total, 60 backlog, 50 overdue, 30/48 = 62.5%, S$35,827.50. |
| Copy | The saved copy still has the .Rules sheet. |

## Troubleshooting

| If this happens | Fix |
|---|---|
| Copilot cannot see my file | In Excel, save the file and open Copilot again. In Word, PowerPoint or Outlook, upload the file with + > Upload images and files, and upload it again after you change it. |
| A number is different from the lab | Clear any slicers or filters, check the table has no extra rows, then ask again in a new chat. |
| A button or feature is missing | Check you are signed in with your course account and Microsoft 365 is up to date, then tell your trainer. |

## Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

## Reflection

Which evidence proves the output is correct, and which decision still needs a person?